26900054 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBTS301 - INTRODUCTION TO FORENSIC BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Evidence is considered relevant when it _____. [K] / [CO4]

   a). Has a logical connection to an issue before the court

   b). Is very old

   c). Is difficult to analyse

   d). Has a high monetary value

2. _____ is a consequence of contamination. [K] / [CO2]

   a). False DNA matches b). Better DNA quality

   c). Increased sample quantity d). Faster DNA extraction

3. Sex estimation becomes less reliable when _____. [K] / [CO5]

   a). The pelvis is well preserved

b). Complete skeleton is available

c). Only small skeletal fragments are available

d). Population standards are available

4. Confirmatory analysis of blood may include _____. [K] / [CO2]

a). DNA profiling b). Luminol only

c). Luminol only d). Kastle–Meyer only

5. The modern forensic method of DNA profiling mainly uses _____. [K] / [CO3]

a). RFLP b). STR analysis

c). RNA analysis d). Protein profiling

6. Mitochondrial DNA is inherited mainly through the _____. [K] / [CO2]

a). Father b). Mother

c). Both parents equally d). Grandfather

7. Which analysis determines the amount or concentration of a substance? [K] / [CO3]

a). Qualitative analysis b). Microscopic analysis

c). Quantitative analysis d). Physical analysis

8. Which of the following should be recorded when assessing skeletal completeness? [K] / [CO1]

a). Missing bones b). Blood group

c). Blood pressure d). Drug concentration

9. During which stage does the opposing side question the expert witness? [K] / [CO4]

a). Examination-in-chief b). Cross-examination

c). Re-examination d). Investigation

10. Which analytical technique is mainly used for volatile and thermally stable compounds? [K] / [CO2]

a). HPLC b). GC

c). TLC d). Immunoassay

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. Why is DNA profiling used in saliva evidence analysis? [U] / [CO2]

12. Define forensic chemistry. [K] / [CO3]

13. What is meant by the responsibility of a forensic scientist to truth? [K] / [CO4]

14. State the major benefits and applications of Raman spectroscopy. [U] / [CO5]

15. Define forensic toxicology. [K] / [CO2]

16. What is digital forensics? Mention two examples of digital evidence. [K] / [CO1]

17. List any three applications of IR spectroscopy in forensic science. [K] / [CO5]

18. Why is age estimation different in juveniles and adults? [U] / [CO5]

19. List any three effects of drugs on the central nervous system. [K] / [CO3]

20. Define chain of custody. [K] / [CO4]

21. What is the principle of microscopic detection of sperm? [U] / [CO2]

22. Who was Edmund Locard? Mention his major contribution. [U] / [CO1]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Discuss the anatomical method of stature estimation from skeletal remains. [U] / [CO5]

24. Describe the basic workflow of mass spectrometry from sample introduction to generation of the mass spectrum. [U] / [CO5]

25. Describe the major stages involved in the interpretation of biological evidence. [U] / [CO3]

26. Discuss the principle and applications of gas chromatography. [U] / [CO5]

27. Discuss the major objectives of forensic anthropology in the examination of human skeletal remains. [U] / [CO1]

28. Describe the workflow involved in forensic drug identification from sample collection to reporting. [U] / [CO5]

29. Discuss the major ethical considerations involved in the use of DNA evidence. [U] / [CO4]

30. Describe the different types of biological evidence and their forensic significance. [U] / [CO3]

[5x6 = 30]